

Problem

A two-port is described by the following equations:

$$\begin{aligned} V_1 &= 50I_1 + 10I_2 \\ V_2 &= 30I_1 + 20I_2 \end{aligned}$$

Which of the following is *not* true?

- (a) $z_{12} = 10$
- (b) $y_{12} = -0.0143$
- (c) $h_{12} = 0.5$
- (d) $A = 50$

Step-by-step solution

Step 1 of 5

$$\begin{aligned} V_1 &= 50I_1 + 10I_2 \\ V_2 &= 30I_1 + 20I_2 \end{aligned}$$

Step 2 of 5

$$(a) \quad Z_{12} = \left. \frac{V_1}{I_2} \right|_{V_1=0} = 10$$

Step 3 of 5

$$\begin{aligned} (b) \quad Y_{12} &= \left. \frac{I_1}{V_2} \right|_{V_1=0} \\ V_2 = 0 &\Rightarrow 30I_1 + 20I_2 = 0 \Rightarrow I_2 = -1.5I_1 \\ \Rightarrow Y_{12} &= \frac{I_1}{-10I_1} = -\frac{1}{10} = -0.1 \end{aligned}$$

Step 4 of 5

$$(c) \quad h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} = \frac{10I_2}{20I_2} = 0.5$$

Step 5 of 5

$$(d) \ A = \left. \frac{V_1}{V_2} \right|_{I_2=0} = \frac{50I_1}{30I_1} = 1.67$$

so, (d) is Not True